

Topology

METRIC SPACE

SET equipped with notion of DISTANCE

ENSEMBLE of n -TUPLES

measured by a METRIC

$$n=3 \quad \left\{ \begin{matrix} x_1 \\ y_1 \\ z_1 \end{matrix} \right\} \quad \left\{ \begin{matrix} x_2 \\ y_2 \\ z_2 \end{matrix} \right\} \quad \dots$$

in $\mathbb{R}^3$ $\uparrow$ p_1 $\uparrow$ p_2

EUCLIDEAN DISTANCE

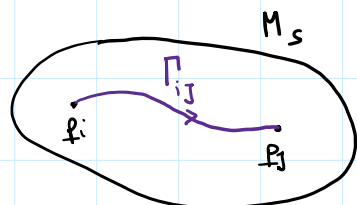
$$n=3 \quad d_{1,2} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Example of metric space $\Rightarrow$ EUCLIDEAN SPACE ($\mathbb{R}^3$)

CONNECTED METRIC SPACE : M_S

$\forall$ two points $p_i, p_j \in M_S$, M_S is connected if there exist Γ_{ij} connecting i, j where $\Gamma_{ij} \in M_S$

every point along Γ_{ij} is within M_S



M_S IS CONNECTED



$\Gamma_{ij} \notin M_S' \Rightarrow M_S'$ is NOT connected

$$M_S' = M_S'' \cup M_S'''$$

DOMAIN :

CONNECTED SUBSET of a METRIC SPACE

connected subset of $\mathbb{R}^3$

$\mathbb{R}^3$

- M_S, π_S'', π_S''' ARE DOMAINS
- M_S' IS NOT a DOMAIN

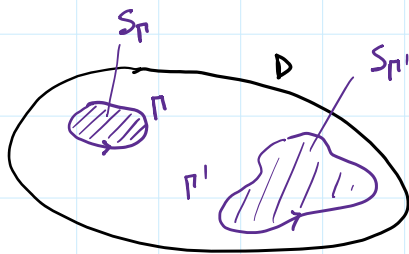
DOMAINS in 2D

SIMPLY-CONNECTED DOMAIN (SCD)

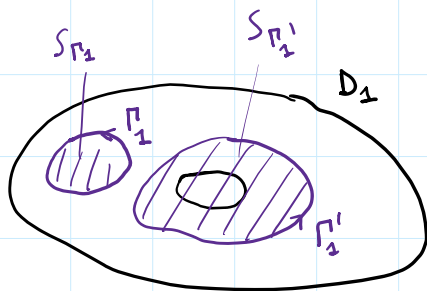
D is SCD if, $\forall \Gamma \in D$ there exist a surface $S_\Gamma \in D$

CLOSED PATH

Surface S_Γ bounded by Γ



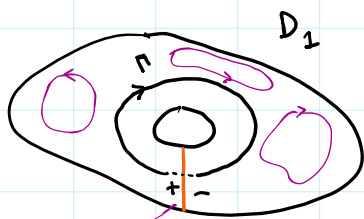
$S_\Gamma \in D$
 $S_{\Gamma'} \in D \Rightarrow D$ is a SCD



$S_{\Gamma_1} \in D_1$
 $S_{\Gamma'_1} \notin D_1 \Rightarrow D$ IS NOT a SCD

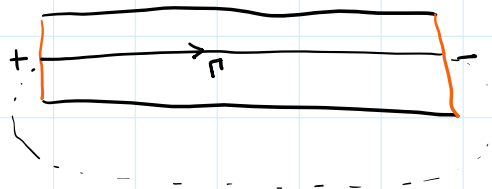
D_1 is MULTIPLY-CONNECTED DOMAIN (MCD)

REDUCTION of MCD to a SCD



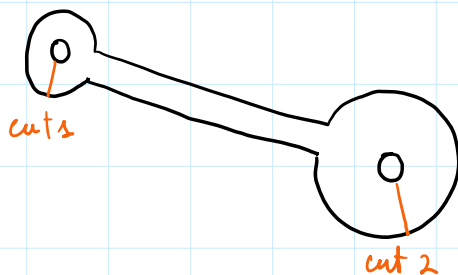
INFINITESIMAL CUT of D_1

$\Gamma \notin D_1 \Rightarrow$ no longer possible to define closed Γ such that $S_\Gamma \notin D_1 \Rightarrow D_1$ reduced to SCD



MULTIPLICITY of a DOMAIN : # of infinitesimal cuts to reduce a MCD to SCD

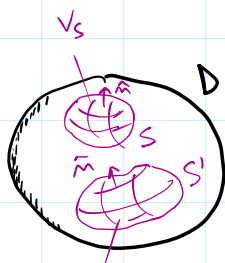
Ex: MCD
wrench



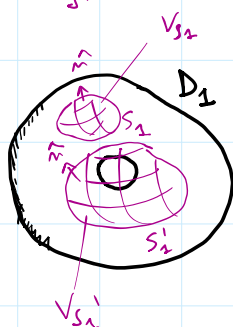
MULTIPLICITY = 2

Domains in 3D

SCD: D is SCD if, $\forall$ closed surfaces $S \in D$ the volume $V_S \in D$
(3D)
↑
bounded by S



$V_S \in D$
 $V_{S'} \in D \Rightarrow$ SCD



$V_{S_2} \in D$
 $V_{S_2'} \notin D \Rightarrow$ MCD (not SCD)

Reduction of MCD (3D)



NOT ABLE TO
define this
closed surface

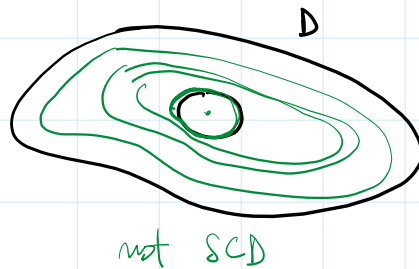
$\Rightarrow D_1$ reduced to
a SCD

Alternative def. of SCD (2D // 3D)

$\hookrightarrow$ domain is SCD if can be SHRUNK to a single

↳ domain is SCD if can be SHRUNK to a single point through a CONTINUOUS transformation

EX:
(2D)



Differential Operators

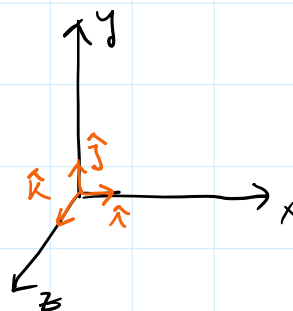
GRADIENT $\mathbb{R} \rightarrow \mathbb{R}^3$

let f be a SCALAR FUNCTION $\rightarrow \nabla f$ describes the partial derivatives of f

Def: $\nabla f \cdot \hat{s} = \frac{\partial f}{\partial s}$
 PROJECTION of f along direction given by $\hat{s}$

in CART. COORDS

$$\nabla f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$



check: $\nabla f \cdot \hat{i} = \frac{\partial f}{\partial x}$

VECTOR field

let $\vec{U} = \nabla f$ $\vec{U}$: VF that can be expressed as grad. of a scalar function

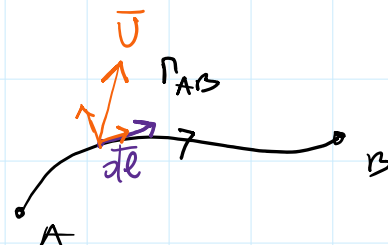
f is a SCALAR POTENTIAL for $\vec{U}$

accumulation of tangential component of the v.f.

LINE INTEGRAL $\vec{U} = \nabla f$

$$\int_{\Gamma_{AB}} \vec{U} \cdot d\vec{\ell} = \int_A^B \underbrace{\nabla f \cdot \hat{\ell}}_{df/d\ell} d\ell =$$

$\nwarrow$ def. of grad
 $\nwarrow$ ∇f
 $\nwarrow$ $\hat{\ell} d\ell$



$$= \int_A^B \underbrace{\frac{df}{d\ell}}_{\text{different of } f} d\ell = \int_A^B df = f(B) - f(A)$$

→ line integral depends only on ENDS of Γ_{AB}

⇒ line integrals of vector fields expressed as gradients of scalar functions are PATH-INDEPENDENT